

```
1  <!--Function with onclick-->
2  <html>
3  <head>
4  <title>A Simple Page</title>
5  <script language="JavaScript">
6  <!--
7  function inputCels()
8  {
9      var cels = prompt("Enter temperature in Celsius:");
10     ansFah = doFahCalc(cels);
11     alert(cels + " Degrees Celsius is " + ansFah + " Degrees Fahrenheit");
12 }
13
14 function inputFah()
15 {
16     var fah = prompt("Enter temperature in Fahrenheit:");
17     ansCel = doCelCalc(fah);
18     alert(fah + " Degrees Fahrenheit is " + ansCel + " Degrees Celsius");
19 }
20
21 function doCelCalc(fah)
22 {
23     var ans = ((Number(fah) - 32) / 1.8);
24     return (ans);
25 }
26
27 function doFahCalc(cels)
28 {
29     var ans = ((1.8 * Number(cels)) + 32);
30     return (ans);
31 }
32 // -->
33 </script>
34 </head>
35 <body>
36 <input type="button" value="Convert Celsius to Fahrenheit" onClick="inputCels();">
37 <br>
38 <br>
39 <input type="button" value="Convert Fahrenheit to Celsius" onClick="inputFah();">
40 </body>
41 </html>
42
```

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1  <!--Function with onsubmit-->
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1  <!--Function & Regular Expression for Credit card validation & onChangeEvent-->
2  <html>
3  <head>
4  <title>Matching a 16 digit credit card number separated by a single space</title>
5  <script type="text/javascript" language="javascript">
6  <!-- //
7  function IsMatchingCard(str){
8      var myRegExp = /[0-9]{4} [0-9]{4} [0-9]{4} [0-9]{4}/;
9      return myRegExp.test(str);
10 }
11
12 function TestGuess(){
13     var cardNumber = document.getElementById("cardNumber").value;
14     if (IsMatchingCard(cardNumber)) {
15         alert("Valid credit card number: " + cardNumber);
16     } else {
17         alert("Invalid credit card number!");
18     }
19 }
20
21 function validateOnChange(){
22     var cardNumber = document.getElementById("cardNumber").value;
23     if (!IsMatchingCard(cardNumber)) {
24         alert("Invalid credit card number!");
25     }
26 }
27 // -->
28 </script>
29 </head>
30 <body>
31 <h3>This page allows you to enter and check a 16 digit credit card number</h3>
32 <form>
33 <input type="text" id="cardNumber" onchange="validateOnChange()" placeholder="Enter
card number">
34 <button type="button" onclick="TestGuess()">Click here to display</button>
35 </form>
36 </body>
37 </html>
38
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1  <!--Function with onClick * Visa Card Regular Expression -->
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1  <!--Function with onClick * Master Card Regular Expression -->
2  <html>
3    <head>
4      <title>MasterCard Example</title>
5      <script type="text/javascript">
6        function luhnCheckSum(sCardNum) {
7
8          var iOddSum = 0;
9          var iEvenSum = 0;
10         var bIsOdd = true;
11
12         for (var i=sCardNum.length-1; i >= 0; i--) {
13
14           var iNum = parseInt(sCardNum.charAt(i));
15
16           if (bIsOdd) {
17             iOddSum += iNum;
18           } else {
19             iNum = iNum * 2;
20             if (iNum > 9) {
21               iNum = eval(iNum.toString().split("").join("+"));
22             }
23             iEvenSum += iNum;
24           }
25
26           bIsOdd = !bIsOdd;
27         }
28
29         return ((iEvenSum + iOddSum) % 10 == 0);
30       }
31
32       function isValidMasterCard(sText) {
33         var reMasterCard =
34           /^[5[1-5]\d{2}) [\s\~]?(\d{4}) [\s\~]?(\d{4}) [\s\~]?(\d{4})$/;
35
36         if (reMasterCard.test(sText)) {
37
38           var sCardNum = RegExp.$1 + RegExp.$2 + RegExp.$3 + RegExp.$4;
39
40           return luhnCheckSum(sCardNum);
41
42         } else {
43           return false;
44         }
45       }
46
47       function validate() {
48         var oInput1 = document.getElementById("txt1");
49         if (isValidMasterCard(oInput1.value)) {
50           alert("Valid");
51         } else {
52           alert("Invalid!");
53         }
54       }
55     </script>
56   </head>
57   <body>
58
59     <P>MasterCard Number: <input type="text" id="txt1" /><br />
60     <input type="button" value="Validate" onclick="validate()" /></p>
61
62   </body>
63 </html>

```